

Indian Institute of Engineering Science and Technology, Shibpur
BTech 3rd Semester End-Semester Examination, November, 2023

Subject: Digital Logic (CS 2102)

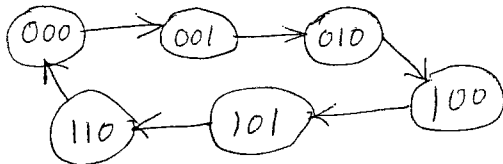
Full Marks: 50

Time: 3 Hours

(Answer question number 1 and any five from the rest)

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1. (i) Give four examples of positional number system
(ii) Convert decimal 153.513 in octal
(iii) What is the use of correction logic in BCD adder
(iv) What are the different way to represent a negative number
(v) Compute $(2^0 - 2^{-4}) - (0.0110)$
(vi) Why BCD is less efficient than binary
(vii) What is the gray code representation of decimal 10 and decimal 15
(viii) Write one advantage and one disadvantage of using CMOS logic family over TTL logic family
(ix) Define fan out and propagation delay time in the context of digital integrated circuit
(x) What is the (8 4 -2 -1) and (2 4 2 1) representation of decimal 9 [2x10]

2. The state diagram of a synchronous counter is shown below. Design the corresponding self-correcting counter using J-K flip-flop. [6]



3. Show the characteristic table, characteristic equation and state diagram of the following flip-flops.
(i) S-R (ii) D (iii) J-K (iv) T [1.5x4]

- 4.(i) Compare the performance of ripple carry adder and carry look-ahead adder in terms of delay and hardware complexity.
(ii) Draw the block diagram of a 4-bit adder-subtractor. [4+2]

- 5.(i) What is the logic of designing 2-bit comparator using two 1-bit comparator?
(ii) How the 8:1 MUX can be used to generate sum and carry of full adder? [2+4]

- 6.(i) Define prime implicant and essential prime implicant.
(ii) Consider the four variable Boolean function $F(A, B, C, D) = \sum(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$. Show all the prime and essential prime implicants. [2+4]

- 7.(i) Boolean function $F = A'C + A'B + AB'C + BC$. Find the minimal SOP expression of F.
(ii) Show all the possible simplified form of the Boolean function $F(W, X, Y, Z) = \sum(1, 3, 7, 11, 15) + \sum_d(0, 2, 5)$. [2+4]

- 8.(i) Boolean function $F(x, y, z) = (x' + y + z)(x' + y + z')(x' + y' + z')$. Show F in simplified SOP form.
(ii) $F1 = x'yz' + x'y'z$ and $F2 = x(y'z' + yz)$. Find $F1'$ and $F2'$ using duality. [2+4]